

The Observer–Context Category: Formal Architecture

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Abstract

Paper D introduces an “observer” as an isometric enlargement of a system together with a measurement context on the enlarged algebra, with two arrow classes: invertible *switches* (context changes at a fixed cut) and non-invertible *refinements* (moving the Heisenberg cut, $A \mapsto A \otimes I$). The non-invertibility of refinements is what forced the correction from “groupoid” to “category.” This note supplies the missing architecture. We argue that the correct home is a *Grothendieck opfibration* $\pi : \mathbb{O} \rightarrow \mathcal{E}$ over the category $\mathcal{E}$ of enlargements — equivalently a pseudofunctor $\mathbf{Ctx} : \mathcal{E} \rightarrow \mathbf{Gpd}$ whose fibers are the switch groupoids and whose reindexing functors are the adjoin-and-tensor maps. This choice (i) subsumes the “category with a wide switch subgroupoid” picture (the subgroupoid is exactly the union of fibers), (ii) is more faithful than a bare double category because it builds the invertibility asymmetry into the variance rather than positing it, and (iii) makes “the cocycle is pulled along a functor” literal. We specify objects, all arrow types with sources and targets, composition, identities, and associativity; prove the switches form a wide subgroupoid; give refinement functoriality on contexts, on the $U(1)$ cocycle, on value assignments, and on obstruction classes with pullback-monotonicity; locate the $U(1)$ cocycle precisely (fiberwise, on the switch subgroupoid, as a reindexing-compatible section); define holonomy only for invertible switch loops and separate it from pullback-along-embeddings; and restate the movable-cut result as an exact five-level statement with a commutative diagram. Every claim is labeled [COMPLETE] complete, [SKETCH] a sketch relying on a named categorical result, or [GAP] a gap. The Copenhagen scope is stated explicitly: the formalism *supports* a Copenhagen-compatible reading; it does not prove Copenhagen is the unique interpretation.

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1 The problem, and the choice of structure

The data. Fix the finite Weyl model: a system Hilbert space $\mathcal{H}$ of dimension d^m (m registers of local dimension d), Weyl operators $W(v)$ indexed by $v \in V = (\mathbb{Z}_d)^{2m}$, with the section law

$$W(u)W(v) = \tau^{c(u,v)}W(u+v), \quad \tau = e^{i\pi/d}, \quad (1)$$

where $c : V \times V \rightarrow \mathbb{Z}_{2d}$ is the canonical 2-cocycle of the companion papers. A *context* is a closed commuting Weyl family (an isotropic subgroup $L \subseteq V$ with its operator family); it need not be maximal. An *observer* adjoins ancillary degrees of freedom (a friend, a memory, an apparatus) by an isometry $\iota : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{H}_O$ and picks a context on the enlarged algebra.

Two arrow classes, one asymmetry. Paper D generates morphisms from two kinds of move:

- (a) *Switches*: at a *fixed* enlargement $\mathcal{K}$, a unitary U on $\mathcal{K}$ carrying a context C to a context C' (a basis change, a Wigner rotation, the friend performing her measurement). Switches are invertible ($U^{-1} = U^*$).
- (b) *Refinements*: *changing* the enlargement, $\iota \rightsquigarrow \iota' \circ \iota$, and embedding observables by $A \mapsto A \otimes I$ (moving the Heisenberg cut outward, from friend to Wigner). Because an isometry into a strictly larger Hilbert space has no two-sided inverse, refinements are *not* invertible.

The single structural fact that organizes everything below is that these two classes *differ in invertibility*. Any faithful formalism must encode that asymmetry, not merely tolerate it.

Three candidate structures.

- (1) **A category with a wide subgroupoid of switches.** Minimal and unimpeachable: declare $\mathbb{O}$ a category, observe that the switches are isomorphisms and form a wide subgroupoid, and take all holonomy invariants there. This is exactly the corrected Paper D wording. It is *true* but under-informative: it says only that some arrows fail to be invertible, without organizing the refinement direction.
- (2) **A double category.** Treat switches and refinements as two 1-cell directions (say vertical and horizontal) with 2-cells the commuting squares “refine-then-switch = switch-then-refine.” This records the compatibility squares explicitly, and is the natural home for the naturality statements. But a bare double category posits the two directions *symmetrically*; the invertibility asymmetry (switches invertible, refinements not) has to be added by hand, and one owes the interchange law and a filling of all squares.
- (3) **A category (op)fibered over enlargements / a pseudofunctor.** Let the base $\mathcal{E}$ be the category of enlargements, let each fiber be the switch groupoid of a fixed enlargement, and let refinements be the (op)cartesian lifts of enlargement maps. Equivalently, a pseudofunctor $\text{Ctx} : \mathcal{E} \rightarrow \mathbf{Gpd}$, $\mathcal{K} \mapsto \text{Sw}(\mathcal{K})$, $e \mapsto (- \otimes I)$.

Our choice, and why. We adopt structure (3): $\mathbb{O}$ is the Grothendieck opfibration of a pseudofunctor $\text{Ctx} : \mathcal{E} \rightarrow \mathbf{Gpd}$ into groupoids. The reasons are exactly the three desiderata of the correction:

- It encodes the invertibility asymmetry *intrinsically*: the invertible switches are precisely the *fiber* (vertical) arrows; the non-invertible refinements are precisely the *cartesian* (horizontal) arrows over non-invertible base maps. Invertibility is not posited; it is the statement “fibers are groupoids.”
- It subsumes structure (1): the wide switch subgroupoid is literally $\bigsqcup_{\mathcal{K}} \text{Sw}(\mathcal{K})$, the union of the fibers, i.e. the vertical subcategory (Theorem 2).
- It makes “the $U(1)$ cocycle is pulled along a functor” literal: reindexing $(- \otimes I)$ is a functor between fibers, and the cocycle-on-the-nose lemma (Lemma 2) is exactly the statement that the fiberwise cocycles form a reindexing-compatible section (§7).

Structure (2) is not discarded: the fibration *induces* a double category (its “framed” / fibrant double category in the sense of Shulman [6]), whose squares are the naturality squares. We use those squares in the movable-cut diagram (§9) but do not take the double category as primitive, because it would add coherence obligations (interchange) without adding faithfulness. We record this relationship honestly in the status table (§11).

Convention (Strictification). To keep composition strictly associative we fix, once and for all, a strictly monoidal model of enlargements: ancilla registers are indexed by a well-ordered set and $\otimes$ is the ordered tensor, so that $(- \otimes I_{O'}) \circ (- \otimes I_O) = (- \otimes I_{O \sqcup O'})$ *on the nose*. Under this convention Ctx is a *strict* 2-functor and the Grothendieck construction is a strict opfibration. Dropping the convention costs only the coherence isomorphisms of a pseudofunctor, and the passage to a fibration is then the standard Grothendieck correspondence [3, 4, 5]; we flag exactly where this is invoked.

2 The base: the category of enlargements $\mathcal{E}$

Definition 1 (Enlargements). The *category of enlargements* $\mathcal{E}$ has:

- **Objects:** cut data, i.e. isometries $\iota : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{H}_O$ with $\mathcal{H}$ the fixed system and $\mathcal{H}_O$ a (possibly trivial) ancilla; write the enlarged space as $\mathcal{K}_\iota := \mathcal{H} \otimes \mathcal{H}_O$. The trivial enlargement $\iota_0 = \text{id}_{\mathcal{H}}$ is an object.
- **Morphisms** $\iota \rightarrow \iota'$: ancilla-adjointing isometries $e : \mathcal{K}_\iota \rightarrow \mathcal{K}_{\iota'}$ of the form $e = \text{id}_{\mathcal{K}_\iota} \otimes \eta$, with $\eta : \mathbb{C} \rightarrow \mathcal{H}_{O'}$ a unit vector (adjoining a fresh ancilla in a fixed state), such that $\iota' = e \circ \iota$. Composition is composition of isometries; the identity is $\text{id}_{\mathcal{K}_\iota}$.

At the level of observables, e induces the unital $*$ -embedding $e_{\sharp} : \mathcal{B}(\mathcal{K}_\iota) \rightarrow \mathcal{B}(\mathcal{K}_{\iota'})$, $A \mapsto A \otimes I$ (Heisenberg picture).

Proposition 1 ($\mathcal{E}$ is a category; refinements are non-invertible). *$\mathcal{E}$ is a well-defined category. Under Convention 1 it is strictly associative and strictly monoidal under $\otimes$. A morphism $e : \iota \rightarrow \iota'$ is invertible in $\mathcal{E}$ iff $\dim \mathcal{K}_\iota = \dim \mathcal{K}_{\iota'}$, i.e. iff the adjoined ancilla is trivial. [COMPLETE]*

Proof. Composition of isometries is an isometry and is associative; identities act as identities; this is a category. Strictness is Convention 1. If $\dim \mathcal{K}_{\iota'} > \dim \mathcal{K}_\iota$ then e is a non-surjective isometry, hence has no right inverse in $\mathcal{E}$ (any r with $e \circ r = \text{id}$ would force e surjective); if dimensions agree, e is a unitary and e^* is its inverse. $\square$

The trivial enlargement ι_0 is initial in the full subcategory on a fixed system $\mathcal{H}$; the whole “moving the cut outward” picture is the poset-like flow away from ι_0 .

3 The fibers: switch groupoids $\text{Sw}(\mathcal{K})$

Definition 2 (Switch groupoid). For a fixed enlarged space $\mathcal{K}$, the *switch groupoid* $\text{Sw}(\mathcal{K})$ has:

- **Objects:** contexts C on $\mathcal{B}(\mathcal{K})$ (closed commuting Weyl families of the enlarged Weyl system).
- **Morphisms** $C \rightarrow C'$: gauge classes $[U]$ of context-carrying unitaries, i.e. U in the Weyl normalizer (Clifford group) with $UCU^* = C'$, taken modulo the *Weyl-section gauge* $\Gamma = \{\tau^k W(a)\}$ acting by $U \mapsto g_2 U g_1$, $g_1, g_2 \in \Gamma$.
- **Composition:** $[U_2] \circ [U_1] = [U_2 U_1]$; **identity:** $\text{id}_C = [I]$.

Proposition 2 ($\text{Sw}(\mathcal{K})$ is a groupoid). $\text{Sw}(\mathcal{K})$ is a well-defined groupoid: composition is associative and gauge-well-defined, and every morphism $[U] : C \rightarrow C'$ has inverse $[U^*] : C' \rightarrow C$. [COMPLETE]

Proof. Matrix multiplication is associative. Gauge-invariance of composition: if $U_1 \sim g_2 U_1 g_1$ and $U_2 \sim h_2 U_2 h_1$, then $U_2 U_1 \sim h_2 U_2 (h_1 g_2) U_1 g_1$; since Γ is a group normalized by the Clifford group, $h_1 g_2$ is absorbed into the left/right gauge of the product, so $[U_2 U_1]$ depends only on $[U_1], [U_2]$. If $UCU^* = C'$ then $U^* C' U = C$, and $[U^*][U] = [U^* U] = [I] = \text{id}_C$, $[U][U^*] = \text{id}_{C'}$. Hence every arrow is invertible. $\square$

Remark (Why the gauge quotient). The gauge Γ is the phase/Weyl ambiguity of the section (1). Quotienting by it makes objects “contexts” rather than “sectioned contexts,” and is exactly the freedom under which the raw holonomy hol is defined only up to sign while hol^2 is invariant (§8). The $U(1)$ cocycle lives on this groupoid as a central extension datum recording that quotient (§7).

4 The total category $\mathbb{O}$ as a Grothendieck opfibration

Construction 1 (The reindexing pseudofunctor). Define $\text{Ctx} : \mathcal{E} \rightarrow \mathbf{Gpd}$ on objects by $\text{Ctx}(\iota) = \text{Sw}(\mathcal{K}_\iota)$ and on a refinement $e : \iota \rightarrow \iota'$ by the *adjoin-and-tensor* functor

$$e_! := (- \otimes I) : \text{Sw}(\mathcal{K}_\iota) \rightarrow \text{Sw}(\mathcal{K}_{\iota'}), \quad C \mapsto C \otimes I, \quad [U] \mapsto [U \otimes I].$$

Lemma 1 ($e_!$ is a functor of groupoids). $e_!$ is a well-defined, faithful functor. Under Convention 1, $(e' \circ e)_! = e'_! \circ e_!$ and $\text{id}_! = \text{id}$, so Ctx is a strict 2-functor. [COMPLETE]

Proof. $U \otimes I$ conjugates $C \otimes I$ to $C' \otimes I$; $(U_2 U_1) \otimes I = (U_2 \otimes I)(U_1 \otimes I)$ and $I \otimes I = I$, so $e_!$ preserves composition and identities. It respects the gauge because $g \otimes I \in \Gamma_{\mathcal{K}'}$ for $g \in \Gamma_{\mathcal{K}}$. Faithfulness: $U \otimes I = U' \otimes I \pmod{\text{gauge}}$ forces $[U] = [U']$. Strict functoriality in e is the definition of the ordered tensor (Convention 1). $\square$

Definition 3 (The observer–context category). $\mathbb{O} := \int_{\mathcal{E}} \text{Ctx}$ is the Grothendieck construction of Ctx :

- **Objects:** pairs $o = (\iota, C)$ with $\iota \in \mathcal{E}$ and $C \in \text{Sw}(\mathcal{K}_\iota)$ a context on the enlarged algebra — exactly Paper D’s observers.

- **Morphisms** $o = (\iota, C) \rightarrow o' = (\iota', C')$: pairs $(e, [U])$ with $e : \iota \rightarrow \iota'$ in $\mathcal{E}$ and $[U] : e_!(C) = C \otimes I \rightarrow C'$ in $\text{Sw}(\mathcal{K}_{\iota'})$.

- **Source/target**: $\text{src}(e, [U]) = (\iota, C)$, $\text{tgt}(e, [U]) = (\iota', C')$.

- **Composition**: for $(e, [U]) : o \rightarrow o'$ and $(e', [U']) : o' \rightarrow o''$,

$$(e', [U']) \circ (e, [U]) := (e' \circ e, [U'] \circ e'_! [U]) = (e' \circ e, [U'] (U \otimes I')). \quad (2)$$

- **Identity**: $\text{id}_{(\iota, C)} = (\text{id}_{\iota}, [I])$.

Theorem 1 ($\mathbb{O}$ is a strict opfibration). *Under Convention 1, $\mathbb{O}$ is a category, the projection $\pi : \mathbb{O} \rightarrow \mathcal{E}$, $(\iota, C) \mapsto \iota$, $(e, [U]) \mapsto e$, is a functor, and π is a split Grothendieck opfibration whose fiber over ι is $\text{Sw}(\mathcal{K}_{\iota})$. Composition (2) is associative and unital. [COMPLETE] (associativity, units, opfibration split structure by direct verification); the identification with the abstract Grothendieck construction, and the pseudofunctor version without Convention 1, is [SKETCH] [3, 4, 5].*

Proof. Unitality. $(\text{id}_{\iota'}, [I]) \circ (e, [U]) = (e, [I] \circ \text{id}_! [U]) = (e, [U])$ and $(e, [U]) \circ (\text{id}_{\iota}, [I]) = (e, [U] \circ e_! [I]) = (e, [U])$ since $e_! [I] = [I]$.

Associativity. Take $(e, [U]) : o_0 \rightarrow o_1$, $(e', [U']) : o_1 \rightarrow o_2$, $(e'', [U'']) : o_2 \rightarrow o_3$. Composing the first two then the third gives base component $e'' \circ (e' \circ e)$ and fiber component

$$[U''] \circ e''_! ([U'] \circ e'_! [U]) = [U''] \circ e''_! [U'] \circ e''_! e'_! [U],$$

using functoriality of $e''_!$ (Lemma 1). Composing the last two first gives base $(e'' \circ e') \circ e$ and fiber $[U''] \circ e''_! [U'] \circ (e'' \circ e')_! [U]$. By Convention 1, $e'' \circ (e' \circ e) = (e'' \circ e') \circ e$ and $e''_! e'_! = (e'' \circ e')_!$, so the two agree. Hence composition is associative.

Functoriality and fibers. π evidently preserves composition and identities, and $\pi^{-1}(\iota)$ (arrows over id_{ι}) is $\text{Sw}(\mathcal{K}_{\iota})$.

Opcartesian lifts. For a base map $e : \iota \rightarrow \iota'$ and object (ι, C) , the arrow $\rho_e := (e, [I]) : (\iota, C) \rightarrow (\iota', e_! C)$ is opcartesian: any $(e' \circ e, [V]) : (\iota, C) \rightarrow (\iota'', C'')$ factors uniquely through ρ_e as $(e', [V]) \circ \rho_e$. The assignment $e \mapsto \rho_e$ is a cleavage, split because $_!$ is strictly functorial. This is the Grothendieck construction of a strict 2-functor [3, 4]. $\square$

Proposition 3 (Opcartesian–vertical factorization). *Every morphism of $\mathbb{O}$ factors uniquely as a refinement followed by a switch:*

$$(e, [U]) = \underbrace{(\text{id}_{\iota'}, [U])}_{\text{switch}} \circ \underbrace{(e, [I])}_{\text{refinement}}.$$

Refinements are the opcartesian arrows $\rho_e = (e, [I])$; switches are the vertical arrows $(\text{id}_{\iota}, [U])$. [COMPLETE]

Proof. $(\text{id}_{\iota'}, [U]) \circ (e, [I]) = (e, [U] \circ e_! [I]) = (e, [U])$. Uniqueness of the (opcartesian, vertical) factorization is the defining property of an opfibration (Theorem 1). $\square$

Dictionary

Paper D term	Structure in $\mathbb{O}$	Invertible?
observer (ι, C)	object of $\mathbb{O}$ / object of fiber $\text{Sw}(\mathcal{K}_{\iota})$	—
switch $C \rightarrow C'$ (fixed cut)	vertical arrow $(\text{id}_{\iota}, [U])$	yes (fiber = groupoid)
refinement $A \mapsto A \otimes I$	opcartesian arrow $(e, [I])$	no (Prop. 1)
“moving the cut”	travel along the base $\mathcal{E}$	no
switch loop	loop in a single fiber $\text{Sw}(\mathcal{K})$	yes (holonomy defined)

5 Switches form a wide subgroupoid

Theorem 2 (Wide switch subgroupoid). *The vertical arrows $\text{Sw}(\mathbb{O}) := \{(\text{id}_\iota, [U])\}$ form a wide subgroupoid of $\mathbb{O}$: it contains every object, is closed under composition and identities, and every one of its arrows is invertible, with $(\text{id}_\iota, [U])^{-1} = (\text{id}_\iota, [U^*])$. As a category $\text{Sw}(\mathbb{O}) = \bigsqcup_{\iota \in \mathcal{E}} \text{Sw}(\mathcal{K}_\iota)$, the disjoint union of the fibers. [COMPLETE]*

Proof. Verticality is preserved by (2): if both base components are identities so is their composite, and $\text{id}_! = \text{id}$ reduces the fiber component to $[U'] \circ [U]$, composition inside $\text{Sw}(\mathcal{K}_\iota)$. Identities $(\text{id}_\iota, [I])$ are vertical. Invertibility is Proposition 2 transported to $\mathbb{O}$: $(\text{id}_\iota, [U^*]) \circ (\text{id}_\iota, [U]) = (\text{id}_\iota, [U^*U]) = (\text{id}_\iota, [I])$. Wideness holds because every object (ι, C) is in the fiber $\text{Sw}(\mathcal{K}_\iota)$. There are no vertical arrows between different fibers (their base components would differ from id), so the subcategory is the disjoint union of fibers. $\square$

This is precisely candidate structure (1) of §1, now *derived* rather than posited: “category with a wide switch subgroupoid” is the underlying category plus its vertical subcategory. All holonomy invariants of §8 live here.

6 Refinement functoriality

Fix a refinement $e : \iota \rightarrow \iota'$ with reindexing $e_! = (- \otimes I)$. We record its action on each layer of structure.

(R1) On contexts and switches. [COMPLETE] By Lemma 1, $e_! : \text{Sw}(\mathcal{K}_\iota) \rightarrow \text{Sw}(\mathcal{K}_{\iota'})$ is a faithful functor of groupoids: $C \mapsto C \otimes I$, $[U] \mapsto [U \otimes I]$. It is injective on objects and faithful on arrows, but not full (the enlarged algebra has contexts and switches touching the ancilla that are not of the form $- \otimes I$). Thus refinement embeds each observer’s switch groupoid into the larger one.

(R2) On the $U(1)$ cocycle (on the nose). [COMPLETE] Let $c_{\mathcal{K}}$ denote the enlarged Weyl cocycle (1) on $\text{Sw}(\mathcal{K})$.

Lemma 2 (Pullback of the cocycle). *$e_!$ pulls the cocycle back strictly: $e_!^* c_{\mathcal{K}_{\iota'}} = c_{\mathcal{K}_\iota}$. Equivalently $(W(u) \otimes I)(W(v) \otimes I) = \tau^{c(u,v)}(W(u+v) \otimes I)$ with the same exponent $c(u,v)$, for any enlargement. [COMPLETE]*

Proof. $(W(u) \otimes I)(W(v) \otimes I) = W(u)W(v) \otimes I = \tau^{c(u,v)}W(u+v) \otimes I$ by (1); the phase $\tau^{c(u,v)}$ is unchanged, so the cocycle is preserved verbatim. Machine certificate: the six embedded Peres–Mermin context products at $d = 8$ are $(+1)^{\times 5}, (-1)^{\times 1}$, identical to $d = 4$ (`observer_groupoid.py`, part A). $\square$

(R3) On value assignments (pullback–monotonicity). [COMPLETE] A value assignment on a family F of contexts is a section of the presheaf of $\mathbb{Z}_d$ -valued (or $U(1)$ -valued) noncontextual valuations.

Lemma 3 (Restriction and monotonicity). *Refinement induces a restriction map $e_!^*$ on value assignments: a value assignment on $e_!(F) = F \otimes I$ restricts along $e_!$ to one on F . Hence*

F has no global value assignment $\implies$ no enlarged family containing $e_!(F)$ has one.

In particular the Peres–Mermin obstruction survives every Wigner enlargement. [COMPLETE]

Proof. $e_!$ is an injective strict morphism of context structures preserving all operator products (Lemma 2); a valuation ν on a family containing $e_!(F)$ pulls back to $\nu \circ e_!$ on F , which is again noncontextual and consistent because products are preserved on the nose. Contrapositive gives the monotonicity. Certificate as in Lemma 2. $\square$

(R4) On obstruction classes (pullback–monotonicity of the class). [COMPLETE]/[SKETCH] Let $[\theta_F] \in H^2(-; U(1))$ (equivalently the $\mathbb{Z}_d/\text{AvN}$ obstruction of the companion papers) be the obstruction class of a family.

Proposition 4 (Class pullback). $e_!^*[\theta_{e_!(F)}] = [\theta_F]$: the obstruction class of the embedded family equals the pullback of the class upstairs. Consequently the class is monotone under refinement — it cannot be trivialized by enlarging the cut — and $e_!$ never creates or destroys the AvN sector. [COMPLETE] for the equality of representing cocycles (Lemma 2); [SKETCH] for the statement at the level of cohomology classes, which uses functoriality of $H^2(-; U(1))$ under the groupoid functor $e_!$ [9, 10, naturality of group/groupoid cohomology].

Proof. Cocycle representatives are preserved by Lemma 2; naturality of H^2 under $e_!$ turns this into an equality of classes. Monotonicity is the contrapositive together with (R3). $\square$

Remark (Direction of variance). Refinement $e_!$ is *covariant* on contexts/switches ($C \mapsto C \otimes I$) but the induced map on cocycles/classes is *contravariant* (pullback). This is why the natural home is an *opfibration* (covariant reindexing of fibers) carrying a *contravariant* cocycle section — see §7.

7 Where the $U(1)$ cocycle lives

We can now state precisely where the cocycle lives and why.

Definition 4 (Fiberwise cocycle / reindexing-compatible section). For each enlargement ι let $c_{\mathcal{K}_\iota} \in Z^2(\text{Sw}(\mathcal{K}_\iota); U(1))$ be the enlarged Weyl cocycle, i.e. the 2-cocycle of the central extension $U(1) \rightarrow U(\mathcal{K}_\iota) \rightarrow \text{PU}(\mathcal{K}_\iota)$ restricted along the switch groupoid, normalized on Weyl operators by (1). The family $\{c_{\mathcal{K}_\iota}\}_{\iota \in \mathcal{E}}$ is a *reindexing-compatible section* of the contravariant assignment $\iota \mapsto Z^2(\text{Sw}(\mathcal{K}_\iota); U(1))$: by Lemma 2,

$$e_!^* c_{\mathcal{K}_{\iota'}} = c_{\mathcal{K}_\iota} \quad \text{for every refinement } e. \quad (3)$$

Theorem 3 (Location of the cocycle). *The $U(1)$ cocycle lives on the switch subgroupoid, fiberwise, as the reindexing-compatible section (3) — equivalently, as a $U(1)$ -central extension of the vertical subcategory $\text{Sw}(\mathbb{O})$ that is refinement-equivariant. It does not live on the whole category $\mathbb{O}$ as an ordinary group/groupoid 2-cocycle. [COMPLETE] (fiberwise existence and compatibility). The assembly into a single class on all of $\mathbb{O}$ is [GAP] (see below).*

Justification. Why the switch subgroupoid. A $U(1)$ 2-cocycle with holonomy is the datum of a central extension, and holonomy of a loop requires the loop to be *invertible*. Only the vertical (switch) arrows are invertible (Theorem 2); the non-invertible refinements cannot carry holonomy, only pull the cocycle back (Lemma 2). Hence the natural carrier is $\text{Sw}(\mathbb{O}) = \bigsqcup_\iota \text{Sw}(\mathcal{K}_\iota)$, one central extension per fiber.

Why fiberwise-plus-compatible rather than global. A single 2-cocycle on the whole category $\mathbb{O}$ would be an element of a cohomology theory of $\mathbb{O}$ that accepts non-invertible arrows — Baues–Wirsching or Thomason cohomology of the category, i.e. H^2 of the nerve $N\mathbb{O}$ with the appropriate

coefficient system [7, 8]. Ordinary groupoid 2-cocycles are not defined on non-invertible arrows. What we *have* constructed is the next best and arguably correct object: a cocycle on each fiber, glued by the reindexing pullback (3). This is exactly a section of the fibered cohomology of π , and it is what “the cocycle is pulled along a functor” means. $\square$

Remark (The gerbe / nerve viewpoint — [GAP]). Two stronger packagings are natural but not established here. (a) The compatible section (3) is the cocycle data of a $U(1)$ -gerbe over the base $\mathcal{E}$ (a stack of $U(1)$ -central extensions of the switch groupoids); proving it satisfies the gerbe descent/coherence conditions is a [GAP]. (b) A single class in $H^2(N\mathbb{O}; U(1))$ (Baues–Wirsching) restricting to $c_{\mathcal{K}_\iota}$ on each fiber would be the ideal “one cohomological object” the program advertises; constructing it, or proving the obstruction to its existence, is the **main open architectural question** (§11). Everything downstream (holonomy, movable cut) uses only the fiberwise section, which is complete.

8 Holonomy, and its distinction from pullback

Definition 5 (Switch-loop holonomy). Let $\gamma = [U_k] \circ \cdots \circ [U_1]$ be a loop in a single fiber $\text{Sw}(\mathcal{K})$ (so $U_k \cdots U_1$ carries a context to itself). Choose a special-unitary closer W_c and set

$$\text{hol}(\gamma) = \det(W_c U_k \cdots U_1), \quad \text{hol}^2(\gamma) = \det(U_k \cdots U_1)^2.$$

Lemma 4 (Holonomy is a fiber invariant; hol^2 is gauge-robust). *hol is defined for invertible switch loops only. hol itself is section-dependent (changes by ± 1 under Weyl regauging of any switch and by the choice of closer), but $\text{hol}^2 \in \{\pm 1\}$ is invariant under Weyl regauging and closer choice, hence a well-defined $\mathbb{Z}_2$ -valued function on switch loops. [COMPLETE] (Paper D Lemma 2; every Weyl element has $\det = \pm 1$ and closers are special unitary; machine-checked over all 16 Weyl elements, `observer_groupoid.py` part B).*

Proposition 5 (Holonomy vs. pullback are different operations). *Let γ be a switch loop in $\text{Sw}(\mathcal{K}_\iota)$ and $e : \iota \rightarrow \iota'$ a refinement. Then:*

- (i) *Holonomy is the trace of going around an invertible loop inside one fiber; it is a number in $U(1)$ (resp. $\text{hol}^2 \in \mathbb{Z}_2$).*
- (ii) *Pullback along a refinement is the functor $e_!$ between different fibers; a refinement is not a loop and has no holonomy of its own.*
- (iii) *The two interact by preservation: $\text{hol}(e_! \gamma) = \text{hol}(\gamma)$ and $\text{hol}^2(e_! \gamma) = \text{hol}^2(\gamma)$, because $e_!$ preserves the cocycle on the nose (Lemma 2) and $\det(A \otimes I_n) = \det(A)^n$ with the closer scaled accordingly, so the $\mathbb{Z}_2$ residue is unchanged.*

[COMPLETE] (i)–(ii) by definition; (iii) [COMPLETE] for the hol^2 residue, [SKETCH] for the raw hol phase where the ancilla dimension enters $\det(A \otimes I_n) = \det(A)^n$ and one must track the closer normalization.

The upshot, which the “groupoid” error obscured: *moving the cut is not traversing a loop*. Refinement is pullback-along-an-embedding between fibers; it transports holonomy but is not itself a holonomy. Only genuine switch loops — the friend rotating her basis and returning — have holonomy. This is why holonomy is defined solely on the switch subgroupoid.

9 The movable cut, exactly

We restate Paper D’s Theorem 3 as an exact multi-level statement. Fix an observer $o = (\iota, C)$ (the *friend* description), a refinement $e : \iota \rightarrow \iota'$ to $o^\dagger = (\iota', C \otimes I)$ (the *Wigner* description), and a switch $s = [U] : C \rightarrow C'$ in the friend fiber (the friend’s basis change / measurement). Refinement transports s to $e_! s = [U \otimes I] : C \otimes I \rightarrow C' \otimes I$.

Theorem 4 (Movable cut, five separated levels). *With the data above, and any ancilla state σ ($\text{tr } \sigma = 1$):*

- (1) **Empirical-probability invariance** [COMPLETE]. *For every state ρ on $\mathcal{K}_\iota$ and every outcome effect E measured in the switched context,*

$$\text{tr}(\rho E) = \text{tr}((\rho \otimes \sigma)(E \otimes I)).$$

Refinement changes no Born probability; friend and Wigner assign identical statistics.

- (2) **Ray-level (projective) equivalence** [COMPLETE]. *The square below commutes strictly in $\mathbb{O}$ (equivalently at the PU/density-operator level): the comparison map $R = \rho_e$ intertwines the friend and Wigner descriptions, and the ray-level holonomy of the switch loop is +1 (hardware-measured on `ibm_fez`: loop phase $-2.9^\circ \pm 1.6^\circ$).*
- (3) **Section-dependent phase** [COMPLETE]. *Lifting rays to vectors (choosing a determinant section) makes the comparison defined only up to $U(1)$: around a closed switch loop γ the section phase is exactly the holonomy $\text{hol}(\gamma)$ (Definition 5); for the mutually-unbiased triangle $\text{hol} = -i$ in the Clifford section. This phase is bookkeeping, not statistics.*
- (4) **Cohomology-class invariance** [COMPLETE]. *The gauge-robust residue $\text{hol}^2(\gamma) \in \mathbb{Z}_2$ and the obstruction class $[\theta]$ are unchanged by the move and by re-sectioning (Lemma 4, Proposition 4).*
- (5) **Interpretive conclusion — interpretation, not theorem.** *“The cut may be moved without empirical consequence, the section bookkeeping recording the move as a $U(1)$ phase whose invariant residue is the class.” This is the reading the formalism supports; it is labeled interpretation and is not claimed unique (§10).*

Proof. (1) $\text{tr}((\rho \otimes \sigma)(E \otimes I)) = \text{tr}(\rho E) \text{tr}(\sigma) = \text{tr}(\rho E)$. (2) Strict commutativity of the square is Proposition 3/(2): $(\text{id}_{\iota'}, [U \otimes I]) \circ (e, [I]) = (e, [U \otimes I]) = (e, e_![U]) = \rho_e \circ (\text{lift of } s)$ at the level of gauge classes (rays). The measured ray-level loop phase is +1 by Lemma 4 applied at the PU level. (3) Choosing a determinant section is a set-theoretic lift $\text{PU} \rightarrow \text{U}$; its failure to be a functor around γ is precisely $\text{hol}(\gamma)$ (Definition 5), section-dependent by Lemma 4. (4) hol^2 and $[\theta]$ are the gauge-/section-invariants of Lemma 4 and Proposition 4. (5) is flagged as interpretation. $\square$

$$\begin{array}{ccc} (\iota, C) & \xrightarrow{s=[U]} & (\iota, C') \\ R=\rho_e \downarrow & & \downarrow R=\rho_e \\ (\iota', C \otimes I) & \xrightarrow{e_! s=[U \otimes I]} & (\iota', C' \otimes I) \end{array}$$

Reading of the diagram. Horizontal arrows are switches (friend’s move, invertible); vertical arrows are the refinement (moving the cut, non-invertible). **At ray level** (Theorem 4(2)) the square commutes on the nose. **At vector/section level** (Theorem 4(3)) it commutes up to a $U(1)$ phase; **around a closed switch loop** that phase is hol , with invariant residue hol^2 (Theorem 4(4)). The square is the naturality square of the induced double category (Shulman [6]); we use it here only as a picture of the exact statement.

Remark (What is *not* proved). Theorem 4 proves invariance of statistics, of the class, and the exact section-phase bookkeeping for *this* friend/Wigner pair and *these* moves. It does *not* prove the fully general Copenhagen “movable cut principle” for arbitrary system/apparatus splits, nor that the update rule is unique beyond the constrained affine depolarizing family of the companion note (whose unique single-Kraus member is Lüders). Those are the boundaries of the claim.

10 Copenhagen scope

Proposition 6 (Scope, explicit). *The architecture of §§1–9 is Copenhagen-compatible: local classicality per context, complementarity as holonomy of switch loops, relative facts as fiberwise value assignments obstructed globally, and the movable cut as pullback-along-refinement, are all theorems of the stated finite-Weyl axioms. It does not establish that Copenhagen is the unique interpretation consistent with these structures. Everything proved here is available to rival interpretations verbatim; what is fixed is the mathematics (an opfibration with a fiberwise $U(1)$ cocycle), not the metaphysics. [COMPLETE] (as a scope statement).*

Concretely: nothing above modifies a Born-rule prediction; the constructions are reorganizations of finite quantum data. A many-worlds or Bohmian reader may accept Theorems 2–4 without accepting the interpretive gloss 4(5). We therefore present the results as *structural characterizations within the axioms*, not as an interpretation-free derivation of Copenhagen.

11 Status, and the shape of the object

Status table

Ingredient	Status
Base category $\mathcal{E}$ of enlargements; non-invertibility of refinements (Prop. 1)	[COMPLETE]
Fibers $\text{Sw}(\mathcal{K})$ are groupoids (Prop. 2)	[COMPLETE]
Reindexing $e_! = (- \otimes I)$ is a faithful functor; Ctx strict 2-functor (Lem. 1)	[COMPLETE]
Objects, arrows, src/tgt , composition, identities (Def. 3)	[COMPLETE]
Associativity and unitality of $\mathbb{O}$ (Thm. 1)	[COMPLETE]
$\mathbb{O} = \int_{\mathcal{E}} \text{Ctx}$ is a split opfibration (Thm. 1)	[SKETCH] [3, 4, 5]
Pseudofunctor version without strictification (Conv. 1)	[SKETCH] [3, 5]
Switches form a wide subgroupoid $= \bigsqcup_{\iota} \text{Sw}(\mathcal{K}_{\iota})$ (Thm. 2)	[COMPLETE]
Opcartesian-vertical factorization (Prop. 3)	[COMPLETE]
Refinement action on contexts/switches (R1)	[COMPLETE]
Cocycle pulled back on the nose (Lem. 2; certificate $d = 8$)	[COMPLETE]
Value-assignment restriction & monotonicity (Lem. 3)	[COMPLETE]
Obstruction- <i>class</i> pullback (Prop. 4)	[COMPLETE] cocycle / [SKETCH] class [9, 10]
Cocycle located: fiberwise, switch subgroupoid, compatible section (Thm. 3)	[COMPLETE]
Single class on all of $\mathbb{O}$ ($H^2(N\mathbb{O}; \text{U}(1))$, Baues-Wirsching) (Rem. 7)	[GAP]
$\text{U}(1)$ -gerbe over $\mathcal{E}$ packaging (Rem. 7)	[GAP]
Holonomy defined on switch loops; $\text{hol}^2 \in \mathbb{Z}_2$ gauge-robust (Lem. 4)	[COMPLETE] [1]
Holonomy $\neq$ pullback; preservation $\text{hol}(e_! \gamma) = \text{hol}(\gamma)$ (Prop. 5)	[COMPLETE]/[SKETCH]
Movable cut, levels (1)–(4) (Thm. 4)	[COMPLETE]
Movable cut, level (5) interpretive	interpretation
Copenhagen scope statement (Prop. 6)	[COMPLETE]
Induced (framed) double category of squares (Shulman)	[SKETCH] [6]

Is it a category, double category, bicategory, or fibration?

Best judgment. The correct minimal-yet-faithful structure is a **split Grothendieck opfibration** $\pi : \mathbb{O} \rightarrow \mathcal{E}$ **over the category of enlargements, equivalently a (strict, after Convention 1) pseudofunctor** $\text{Ctx} : \mathcal{E} \rightarrow \mathbf{Gpd}$ **into groupoids**, carrying a *fiberwise* $\text{U}(1)$ cocycle compatible under reindexing (3). Concretely:

- It is *not merely* a category: calling it “a category with a wide switch subgroupoid” (candidate 1) is *true* (Theorem 2) but forgets the base direction that makes the refinements organized rather than ad hoc.
- It is *not primitively* a double category or bicategory: the double category (candidate 2, Shulman’s framed double category of the fibration) is *induced*, and its squares are the movable-cut

naturality squares, but taking it as primitive imposes interchange coherence that the fibration gives for free and hides the invertibility asymmetry.

- It *is* a fibration in the honest sense that the invertible/non-invertible split is exactly the vertical/opcartesian split, and the cocycle’s “pulled along a functor” behavior is exactly reindexing.

The main open architectural question. Whether the fiberwise, reindexing-compatible cocycle (3) *assembles into a single cohomology class on all of $\mathbb{O}$* . Precisely: does there exist a class in the Baues–Wirsching / Thomason cohomology $H^2(N\mathbb{O}; U(1))$ (a cohomology theory admitting the non-invertible refinement arrows) restricting to the canonical Weyl cocycle c_{κ_i} on every fiber, and if so is it unique; equivalently, is the compatible section (3) the cocycle datum of a genuine $U(1)$ -gerbe over $\mathcal{E}$? Our results need only the fiberwise section, which is complete; the single-object global class is the natural next target and is currently a [GAP]. A secondary open question is whether the raw phase hol (not merely hol^2) admits a section-free definition once the ancilla-dimension bookkeeping of Proposition 5(iii) is normalized functorially.

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